Department of Computer Science

CS 331: Database System Design and Management

Spring 2026

Credits: 3
Class sessions: Wednesday 8:30 AM – 11:20 AM, GITC 3600

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Prerequisites

- CS 114. Introduction to Computer Science II, or
- CS 116. Introduction to Computer Science II in C++, or
- IT 114. Advanced Programming for Information Technology.

Course Overview

The objective of the course is to provide an introduction to modern database systems. It focuses on the following topics: data models, the entity-relationship model, the relational model, relational algebra, the database language SQL, functional dependencies and normal forms, indexing and concurrent transaction processing.

Students will learn how to design, create, query, and update a database through a number of assignments and a term project. They will acquire hands-on experience with modern database management systems (DBMS) using the standard database language SQL.

Course (learning) outcomes

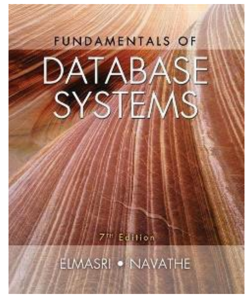
1. Understand the data requirements of contemporary organizations and how database management systems meet them.
2. Develop conceptual data model specifications.
3. Design and implement database applications.
4. Understand how data is stored, retrieved, and maintained in different types of databases.
5. Acquire experience with SQL.

New Jersey Institute of Technology

Textbook

Fundamentals of Database Systems, 7th Edition,
R. Elmasri and S.B. Navathe, Pearson, 2016.
ISBN-10: 0-13-397077-9
ISBN-13: 978-0-13-397077-7

Spotted online here.



Course Outline (tentative)

WEEK	DATE	TOPICS
1	1/21/2026	Database concepts and architecture
2	1/28/2026	The Entity-Relationship (ER) model
3	2/04/2026	Mapping ER models to relational models
4	2/11/2026	Basic SQL
5	2/18/2026	Basic SQL
6	2/25/2026	Advanced SQL
7	3/04/2026	**MIDTERM EXAM**
8	3/11/2026	Advanced SQL
	3/18/2026	**SPRING BREAK**
9	3/25/2026	Relational algebra
10	4/01/2026	Database dependencies and normal forms
11	4/08/2026	Database file organization and indexing
12	4/15/2026	Query processing and optimization
13	4/22/2026	Concurrent transaction processing
14	4/29/2026	Review
	TBD	**FINAL EXAM**

Course Organization

The slides for each lecture will be made available on Canvas before the class. A good practice is to read from the textbook the material to be taught in class and come prepared.

Lecture slides, exercises, homework assignments, and project requirements will be available for download in due course on Canvas. Important announcements will be also placed there. The course Web site is a "living document" that students should visit at least once a week.

Deliverables

1. Three homework assignments consisting of exercises on select topics studied in class. If you put in the work, you will score well. Solutions will be provided to compare with your own. Assignments **must** be done in pairs.
2. A team project to design and implement a simple database system in SQL. A team **must** contain

four students. The project will proceed incrementally in three stages with three deliverables, including a demonstration and report at the end.

3. Two exams: a midterm halfway through the semester, and a final at the end of the course. No Internet access will be allowed during the exams.

Submissions and Late Policy

The homework assignments and all project deliverables should be submitted on or before the time they are due through Canvas. A written submission should be a **SINGLE FILE** in **PDF** format **only**, accompanied by **SQL** source files (in text format) as necessary. Handwritten or hand-drawn submissions will not be accepted. Late submissions will not be accepted or will incur penalties. Only one submission is required per team.

Make-up Exams

Make-up exams need to be approved by the instructor. Consideration will be given to those students who reach out **before** the exam (e-mail/phone) and provide a valid, documented reason for missing the exam.

Grading

The course deliverables contribute to the final course grade (0-100) as follows: Assignments – 10%, Project – 30%, Midterm – 25%, Final – 35%. The final grade will be converted to a letter grade as follows: **A**: 85-100, **B+**:75-84, **B**: 65-74, **C+**: 55-64, **C**: 45-54, **D**: 35-44, **F**: 0-34.

SQL

For homework assignments and the course project, we will use the SQLite command line tool available at https://www.sqlite.com and the SQLiteStudio IDE available at https://www.sqlitestudio.pl.

Many resources are available online. e.g.:
Interactive Online SQL Training
Advanced Online SQL Training

Attendance and Participation

Students are expected to attend all course sessions **and bring a laptop to class**. Participation is highly encouraged to make the class more interactive. Experience shows that students that do not attend classes do not perform well in the midterm and final exams. If you miss a class, be sure to consult with your classmates to fill in the content of the class. Visit the course web page regularly to get notes, assignments, deadlines, and announcements.

Academic Integrity

Academic Integrity is the cornerstone of higher education and is central to the ideals of this course and the university. Cheating is strictly prohibited and devalues the degree that you are working on. As a member of the NJIT community, it is your responsibility to protect your educational investment by knowing and following the Code of Student Conduct and Academic Integrity Policy that is found at: Code of Student Conduct and Academic Integrity Policy.

Please note that we are required to report any suspected academic misconduct to the Office of the Dean of Students. Academic misconduct includes, but is not limited to, cheating, plagiarism, and the inappropriate use of online tools or software. Students who are found responsible for violating the Academic Integrity Code may be subject to disciplinary action. Possible outcomes can include a failing grade on an assignment or in the course, as well as additional sanctions up to and including suspension or dismissal from the university. If you have questions about the Code of Student Conduct and Academic Integrity or the student conduct process, you may contact communitystandards@njit.edu. You may also contact the Dean of Students office at dos@njit.edu

Generative AI

Student use of artificial intelligence (AI) is permitted in this course for assignments and projects. However, if you use AI in this course, the AI tool used must be clearly cited. Please consult the student guide, AI-U to learn more about the use of AI in your studies.

Free Tutoring

Free in-person and remote tutoring and writing assistance is available for students at NJIT, including tutoring in CS 331, offered by the Ying Wu College of Computing. Please consult the YWCC Undergraduate Tutoring Page for details.

Accessibility

We follow the NJIT policies for accommodating and supporting students with disabilities. Please look here for more details.

Religious Observances

NJIT is committed to supporting students observing religious holidays, especially those listed in the NJIT Religious Holiday Calendar, as specified in the NJIT Policy on Student Absences for Religious Observances. Students should complete the accommodation form if there are any conflicts between course requirements and religious observances by the end of the second week of classes and no later than two weeks before the anticipated absence. We will provide academically appropriate accommodation, allowing students to complete missed assignments, exams, quizzes, or other coursework within the term. For questions or additional guidance, please contact the Office of Inclusive Excellence at inclusiveexcellence@njit.edu.

Sexual Misconduct or Harassment

NJIT seeks to maintain a safe learning, living, and working environment free from all types of sex-based and gender-based discrimination prohibited by state and federal laws, including Title IX and Title VII, and in keeping with the university's culture of belonging, values and expected behaviors. NJIT's Title IX Sexual Harassment Policy applies to all members of the university community, including students, faculty, staff, and visitors.